

Bhumika Sachdev — Personal & Professional Profile

About Me

Hi, I'm Bhumika. I have a background in Computer Science and Engineering with a specialization in Data Science, and around two years of professional experience across enterprise applications, automation, data science, Python, and Generative AI.

I have been interested in programming since I was a child. I started learning and working with languages such as C++ and Java, and over time my interests evolved toward data science, artificial intelligence, machine learning, and Generative AI.

I would describe myself as **enthusiastic, stubborn, and ambitious**. When I get stuck on a problem, I tend to keep working on it until I understand and solve it. I am also calm under pressure and enjoy learning new technologies and figuring out how they can be applied to practical problems.

Professional Experience

I currently work at **Accenture** on a project for the **UK Department for Work and Pensions (DWP)**. The project involves modernizing and automating parts of a legacy pension system.

My work has given me exposure to enterprise applications, automation, software development workflows, testing, and increasingly, Generative AI.

One of the AI initiatives I am currently working on involves building a **Retrieval-Augmented Generation (RAG) solution using DeepSeek** to assist with testing different functionalities of the modernized system.

The idea is to use project-specific requirements and technical or functional documentation as the knowledge base. Relevant information is retrieved from these documents and provided as context to an LLM, which can then generate or assist with validating test scenarios.

This approach can automate some of the repetitive work involved in understanding requirements and creating test scenarios. It can also help validate whether the functionality implemented in the modernized system aligns with the original business requirements.

Through this work, I have gained practical exposure to:

- Retrieval-Augmented Generation (RAG)
- Large Language Models (LLMs)
- Embeddings
- Vector search
- Document retrieval
- Prompt engineering
- AI-driven automation
- AI-assisted software testing
- Requirements and documentation-based retrieval

- LLM-based test scenario generation and validation

I also have experience using **GitLab** within my Accenture project environment for managing and making changes to project code.

Previous Experience

As an intern at **Omnivore**, I worked with **SQL and data analysis**, which gave me practical exposure to working with data and developing analytical skills.

This experience, combined with my education and subsequent professional work, helped build my foundation in Python, data science, enterprise applications, and AI/ML.

Education

I hold a Bachelor's degree in **Computer Science and Engineering with a specialization in Data Science** from **Dwarkadas J. Sanghvi College of Engineering (DJ Sanghvi), Mumbai**.

My interest in programming started at a young age. I initially worked with C++ and Java and enjoyed the problem-solving aspect of programming. Over time, I became increasingly interested in data science and artificial intelligence.

My interest in technology was also influenced by my family background, as my father worked as a computer engineer.

Final-Year Project

My final-year project focused on **facial frontal generation using TP-GAN**.

The problem involved situations where only a partially visible or non-frontal view of a person's face was available. The goal was to generate a more complete frontal representation of the face.

The project explored the use of **TP-GAN (Transformation-Power Generative Adversarial Network)** for facial frontal-view generation.

This project gave me early exposure to deep learning, generative models, computer vision, and the practical challenges involved in working with AI systems.

Technical Skills

My strongest technical skill is **Python**.

My current technical knowledge and experience includes:

Programming & Data

- Python
- SQL

- Data analysis
- NumPy
- Pandas

Artificial Intelligence & Machine Learning

- Machine Learning
- Deep Learning
- Generative AI
- Large Language Models
- Retrieval-Augmented Generation (RAG)
- Embeddings
- Vector search
- Prompt engineering
- AI automation
- AI agents

RAG & LLM Technologies

I have worked on personal and professional RAG-based projects and have experimented with:

- ChromaDB
- FAISS
- Hugging Face
- OpenAI APIs
- Gemini APIs
- DeepSeek
- PDF-based knowledge retrieval
- Document-based question answering

Frameworks & Tools

I have experience or introductory knowledge of:

- LangChain
- LangGraph — currently learning
- FastAPI
- Docker
- TensorFlow

- PyTorch
- Git
- GitLab

My current focus is on expanding my understanding of RAG architectures, agentic AI, LLM applications, and production-oriented AI systems.

AI/ML Experience and Current Learning

I consider myself to have a strong foundation in AI/ML while continuing to expand my knowledge across the field.

I have worked on multiple personal AI/ML projects, including:

- PDF-based AI applications
- RAG chatbots
- AI agents
- Document retrieval systems
- LLM-based applications
- AI-assisted automation

One example is exploring agents that can assist with **job-application workflows**, such as helping automate parts of the process of finding and applying for relevant opportunities.

I enjoy learning by building. Rather than only studying AI concepts theoretically, I prefer experimenting with technologies and trying to understand how they can be combined to solve practical problems.

Current AI Focus

My current professional interests are moving increasingly toward:

- Generative AI
- RAG
- Agentic AI
- AI agents
- LLM applications
- AI automation
- Production-oriented AI systems

- AI-assisted software development and testing

I am particularly interested in understanding how AI systems can move beyond simple demonstrations and become reliable, useful applications that solve real business problems.

Career Goals

I am currently targeting roles such as:

- AI Engineer
- ML Engineer
- Generative AI Engineer
- Data Scientist
- Data Analyst

I am especially interested in the intersection of **AI and finance** and **AI and fashion**.

I am also open to **international opportunities**.

My immediate career goal is to deepen my technical knowledge, strengthen my fundamentals, gain more hands-on experience, and move into roles where I can work more deeply on building AI/ML and Generative AI solutions.

I want to develop the ability to build AI systems that are not just technically interesting, but also reliable, scalable, and useful in real-world environments.

What Makes Me a Strong Candidate

One of my strongest qualities is that I am a **fast and persistent learner**.

AI is evolving extremely quickly. Technologies that are considered important today may be replaced or supplemented by something completely different in a few years. RAG and agentic AI are examples of how quickly the field has evolved.

Because of this, I believe adaptability and the ability to learn continuously are extremely important.

I may not know every technology that exists today, but I am confident in my ability to learn new technologies, understand their fundamentals, and apply them to problems.

I am also calm under pressure and persistent when solving problems. Once I become invested in solving a technical problem, I tend to keep working on it until I understand what is going wrong and find a solution.

Working Style

I would describe my working style as persistent, curious, and learning-oriented.

When I encounter a difficult problem, I don't like leaving it unresolved. My stubbornness can actually be useful in this context because I tend to keep investigating until I understand the problem.

I also enjoy exploring new technologies independently and learning through experimentation.

I believe that being a good learner is particularly important in AI because the technology and best practices continue to change rapidly.

Interests Outside Technology

Outside of work, I am interested in:

- Fashion
- Travel
- Photography
- Music
- Reading
- Dancing

I particularly enjoy traveling and prefer **tropical destinations**.

I also love learning about a wide variety of topics. My interests are not limited to technology, and I enjoy exploring subjects simply because I find them interesting.

Hobbies & Personal Interests

Reading

I enjoy reading and am currently reading **The Bell Jar by Sylvia Plath**.

Music

I particularly enjoy **jazz music**.

Dancing

Dancing is one of my hobbies and something I enjoy outside of work.

Coin Collection

One of my more unusual hobbies is collecting **coins from the 1980s from different places around the world**.

I enjoy having a collection of objects that have a history and connection to different parts of the world.

Current Personal Development

At the moment, I am focused on self-improvement and expanding my knowledge.

A significant part of this involves strengthening my fundamentals in AI and ML, experimenting with new technologies, and building projects that help me understand AI systems more deeply.

I am also exploring how AI agents can be applied to everyday tasks and workflows.

I am still figuring out my long-term career and life goals. Right now, my focus is on learning, growing, building useful skills, finding meaningful work, and creating a happy and fulfilling life.

How I Approach Technology

I enjoy understanding technology by actually building with it.

For example, learning about RAG is more interesting to me when I can build a chatbot, connect it to documents, experiment with embeddings and vector databases, and see how retrieval affects the quality of an answer.

Similarly, I am interested in AI agents because they provide an opportunity to move from individual LLM interactions toward systems that can reason through tasks and use tools to accomplish practical objectives.

My goal is to continue moving from learning individual technologies toward understanding how to combine them into complete AI applications.

Public-Facing Chatbot Guidance

If someone asks the chatbot about Bhumika, it should primarily use the information in this knowledge base.

The chatbot should accurately distinguish between:

1. **Professional experience** — technologies and systems Bhumika has worked with professionally.
2. **Personal projects** — technologies she has used while building projects independently.
3. **Current learning** — technologies and concepts she is actively exploring.
4. **Career interests** — roles, industries, and areas she wants to pursue.

The chatbot should not invent additional employers, qualifications, projects, skills, responsibilities, achievements, or personal information that are not present in the knowledge base.

If information is unavailable, it should say that it does not have enough information rather than guessing.

The chatbot should also avoid revealing private information that has not been intentionally included as public-facing information.